COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S2
DATE PERFORMED:	August 3-4, 2026	**DATE SUBMITTED:**	August 4, 2026
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Activity NO. 4.1 **Stacks**

1. PROCEDURE OUTPUT

What procedures did you perform?

ILO A:

```
Stack Empty? 0
```

empty() Operation

Observation: Checks whether the stack container is empty. Returns 0 (false), thus indicating that elements exist in the stack.

```
Stack Size: 3
```

size() Operation (Initial)

Observation: Retrieves the number of elements currently held by the stack.

```
Top Element of the Stack: 15
```

top() Operation (Initial)

Observation: Accesses the topmost element from the stack without removing it.

```
Top Element of the Stack: 8
Stack Size: 2
```

pop() and updated size() + top() operations

Observation: pop() removes the top element (15). top() and size() operations afterwards indicate that the new top element is 8 and that the size has reduced to 2.

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ILO B:

B.1. Stacks using Arrays

```
PS C:\Users\jaime\OneDrive\Documents\HOA\hoa3p1> cd "c:\Users\jaime\OneDrive\Documents\HOA\hoa3p1\" ; if ($?) { g++ hoa42.cpp -o hoa42 } ; if ($?) { .\hoa42 }
Enter number of max elements for new stack: 5

Stack Operations:
1. PUSH, 2. POP, 3. TOP, 4. isEMPTY, 5. DISPLAY, 6. EXIT
Choice: 1
New Value: 4

Stack Operations:
1. PUSH, 2. POP, 3. TOP, 4. isEMPTY, 5. DISPLAY, 6. EXIT
Choice: 5
Stack elements (top to bottom): 4

Stack Operations:
1. PUSH, 2. POP, 3. TOP, 4. isEMPTY, 5. DISPLAY, 6. EXIT
Choice: 3
The element on the top of the stack is 4

Stack Operations:
1. PUSH, 2. POP, 3. TOP, 4. isEMPTY, 5. DISPLAY, 6. EXIT
Choice: 4
Stack is Not Empty

Stack Operations:
1. PUSH, 2. POP, 3. TOP, 4. isEMPTY, 5. DISPLAY, 6. EXIT
Choice: 2
Popping: 4

Stack Operations:
1. PUSH, 2. POP, 3. TOP, 4. isEMPTY, 5. DISPLAY, 6. EXIT
Choice: 5
Stack is Empty.

Stack Operations:
1. PUSH, 2. POP, 3. TOP, 4. isEMPTY, 5. DISPLAY, 6. EXIT
Choice: 6
PS C:\Users\jaime\OneDrive\Documents\HOA\hoa3p1>
```

Output of B.1

B.2. Stacks using Linked Lists

```
PS C:\Users\jaime\OneDrive\Documents\HOA> cd "c:\Users\jaime\OneDrive\Documents\HOA\hoa3p1\" ; if ($?) { g++ hoa42b.cpp -o hoa42b } ; if ($?) { .\hoa42b }
After first PUSH: Top of Stack: 1
After second PUSH: Top of Stack: 5
Displaying stack: Stack elements (top to bottom): 5 1
After first POP: Top of Stack: 1
After second POP: Stack is Empty.
Stack Underflow.
```

Output of B.2

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Expression	Valid? (Y/N)	Output (Console Screenshot)	Analysis
(A+B)+(C-D)	Y	Expression: (A+B)+(C-D) Valid? Y	Every open parenthesis (has its corresponding closed parenthesis) correctly placed. The stack is completely emptied out by the end.

((A+B)+(C-D)	N	Expression: ((A+B)+(C-D) Valid? N	One extra open parenthesis (was pushed at the beginning. Loop terminates here; however, stack is not empty (!s.empty()).
((A+B)+[C-D])	Y	Expression: ((A+B)+[C-D]) Valid? Y	Balancing of nested brackets () and [] is done properly. All closing brackets match the top element of the stack.
((A+B]+[C-D]}	N	Expression: ((A+B]+[C-D]} Valid? N	Mismatch of symbols. The reading of] encounters (at the top of the stack, causing isValid to be set to false.

Internal structure of the stack influences memory management, execution of the operations and easiness of coding and maintenance of the code:

1. **Array-Based Stack:**
    - **Implementation:** Utilizes a block of contiguous memory cells and an index that tracks them (top). Needs manual monitoring of Stack Overflow (if the array is full) and Stack Underflow.
    - **Usage & Memory Impact:** Very memory-efficient implementation of a stack for situations when the size is known and does not change. Its efficiency comes from the lack of need to maintain pointers.
2. **Linked List-Based Stack:**
    - **Implementation:** Allocates separate Node elements on the heap at run-time. Operation push adds an element at the beginning, and pop deletes the head element by manual reassignment of pointers and deletion of the cell.
    - **Usage & Memory Impact:** The stack can grow and shrink as required without risks of stack overflow. Requires additional memory allocation for storing addresses (next) for each element and a little overhead due to dynamic allocations.
3. **C++ Standard Template Library (**std::stack**):**
    - **Implementation:** Acts as a container adaptor that utilizes standard containers such as std::deque and std::vector behind the scenes. Allocation of memory, reallocations, and all pointer manipulation is done automatically by the compiler.

- **Usage & Memory Impact:** Provides a clean and well-optimized way to use the interface without any errors. This is the most optimal way to do software development because memory is effectively abstracted away.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Code assistance in supplementary activity to understand how it can be formulated so that it can help finish my work faster and refine its final code. I also used AI to improve my grammar on forming the observations and conclusion. Finally, I used AI to help me formulate the Tools Analysis**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used AI to help me gain insights on how I would form my supplementary activity. It's because it will help me finish my output faster without simply copy-pasting everything from AI. I did not attend the lab session last week for stacks, thus needing more help from AI in understanding the stacks. Then, I used AI to help me understand the Tools Analysis part on how I would compare the different internal representations. Finally, I used AI to correct my grammar mistakes in the observation and conclusion.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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